

Joha Park, Ph.D.

email: johapark@gmail.com
webpage: johapark.github.io

Computational biologist with 10+ years of experience spanning RNA biology, genomics, spatial omics, and CRISPR-Cas9 gene editing. Proven track record building AI/ML frameworks and scalable bioinformatics pipelines that translate large-scale sequencing and imaging data into therapeutic insights. Author of 8 high-impact publications. Currently driving CRISPR off-target discovery, protein and RNA engineering, and variant effect prediction for genotoxicity derisking at a leading gene therapy company – with hands-on experience across the full development arc from pre-IND to BLA.

CURRENT POSITION

07/2026 – present	Principal Computational Scientist	<i>Intellia Therapeutics</i>
03/2023 – 06/2026	Senior Computational Scientist	
>3 yrs	Computational Sciences Genomic Innovation Intellia Therapeutics, Inc., Cambridge, MA, US	
	Roles	
	• Develop and train deep learning models for predicting CRISPR-Cas9 on-target efficacy and population variants-aware genome-wide off-targets • Predict genetic variant effect using genomic AI models for de-risking genotoxicity • Build scalable, reproducible, and streamlined analysis pipelines for multi-omics • Protein and guide RNA engineering using generative AI to advance a DNA writing platform	

PROFESSIONAL EXPERIENCE

01/2020 – 03/2023	Postdoctoral Associate/Fellow	<i>Massachusetts</i>
3 yrs 3 mos	The Picower Institute for Learning and Memory Institute for Medical Engineering and Science Department of Chemical Engineering Massachusetts Institute of Technology, Cambridge, MA, US	<i>Institute of Technology</i>
	Roles	
	• Develop hydrogel-based tissue clearing and expansion methods for spatial omics • Build a scalable AI/ML framework for 3D light-sheet microscopy • Generate a single-cell mouse brain reference atlas	
03/2019 – 12/2019	Postdoctoral Researcher	<i>Seoul National University</i>
10 months	Center for RNA Research Institute for Basic Science Seoul National University, Seoul, Korea	
	Roles	
	• Lead projects on mRNA poly(A)-tail length regulations • Conduct both wet and dry lab experiments for RNA biology • Develop and analyze custom NGS applications	

EDUCATION AND TRAINING

2012 – 2019	Ph.D. in Biological Sciences (RNA Biology & Bioinformatics) RNA Biology Lab (Advisor: V. Narry Kim) Dissertation: “Poly(A) length regulation: deadenylases and the poly(A) barricade”	<i>Seoul National University</i>
2015 – 2018	Expert Research Personnel (Compulsory Military Service)	<i>Republic of Korea Army</i>
2008 – 2012	B.S. in Biological Sciences (<i>cum laude</i>)	<i>Seoul National University</i>

PUBLICATIONS

*: co-first author, #: co-corresponding author

- 1 **J. Park***, M. Kim*, H. Yi*, K. Baeg*, Y. Choi, Y.-s. Lee, J. Lim, V. N. Kim (2023) “Short poly(A) tails are protected from deadenylation by the LARP1-PABP complex” **Nat. Struct. Mol. Biol.**, 30(3):330–338
- 2 **J. Park***, S. Khan*, D. H. Yun*, T. Ku, K. L. Villa, J. E. Lee, Q. Zhang, G. Feng, J. Park, E. Nedivi#, K. Chung# (2021) “Epitope-preserving magnified analysis of proteome (eMAP)” **Sci. Adv.** 7(16):eabf6589
- 3 Y. Kim*#, **J. Park***, S. Kim*, M. Kim, M.-G. Kang, C. Kwak, M. Kang, B. Kim, H.-W. Rhee, and V. N. Kim# (2018) “PKR senses nuclear and mitochondrial signals by interacting with endogenous double-stranded RNAs” **Mol. Cell**, 71(6):1051–1063.e6
- Selected as “Research Highlights” in **Nat. Chem. Biol.**, 14(11):989
- 4 H. Yi*, **J. Park***, M. Ha, J. Lim, H. Chang and V. N. Kim (2018) “PABP cooperates with the CCR4-NOT complex to promote mRNA deadenylation and block precocious decay” **Mol. Cell**, 70(6):1081–1088.e5
- Previewed in **Mol. Cell**, 70(6):987–988
- 5 T. A. Nguyen*#, **J. Park***, T. L. Dang, Y.-G. Choi, and V. N. Kim# (2018) “Microprocessor depends on hemin to recognize the apical loop of primary microRNA” **Nucleic Acids Res.**, 46(11):5726–5736
- 6 K. T. You, **J. Park**, and V. N. Kim (2015) “Role of the small subunit processome in the maintenance of pluripotent stem cells” **Genes Dev.**, 29(19):2004–2009
- 7 T. A. Nguyen, M. H. Jo, Y.-G. Choi, **J. Park**, S. C. Kwon, S. Hohng, V. N. Kim# and J.-S. Woo# (2015) “Functional anatomy of the human Microprocessor” **Cell**, 161(6):1374–1387
- 8 Y.-K. Kim, G. Wee, **J. Park**, J. Kim, D. Baek, J.-S. Kim#, and V. N. Kim# (2013) “TALEN-based knockout library for human microRNAs” **Nat. Struct. Mol. Biol.**, 20(12):1458–1464

Links: [Google Scholar](#), [PubMed](#)

HONORS AND AWARDS

- | | |
|-------------|---|
| 2022 – 2023 | Picower Postdoctoral Fellows, <i>The Picower Institute for Learning and Memory, MIT</i> |
| 2021 | AKN Outstanding Research Awards, <i>Association of Korean Neuroscientists</i> |
| 2019 | Conference Scholarship, <i>Keystone Symposia: Small Regulatory RNAs</i> |
| 2019 | Best Dissertation Awards, <i>School of Biological Sciences, SNU</i> |

2008 – 2012 Presidential Science Scholarship, *Korea Science and Engineering Foundation*

INTERNATIONAL CONFERENCES

- 2019 Cell Symposia: Regulatory RNAs — **short talk**
“Poly(A) barricade as a rate-limiting factor that shapes poly(A) tail length”
- 2018 CSHL Meeting: Regulatory & Noncoding RNAs — **short talk**
“PKR senses nuclear and mitochondrial signals by interacting with endogenous dsRNAs”
- 2017 CSHL Meeting: Eukaryotic mRNA Processing — **short talk**
“Genome-wide evaluation of the role and specificity of deadenylases”
- 2016 EMBL Symposium: The Complex Life of mRNA — *poster*
- 2016 IMBA: 11th Microsymposium on Small RNAs — *poster*
- 2014 Keystone Symposia: RNA Silencing — *poster*

INVITED TALKS

- 08/2023 Special Seminar – *Graduate School of Medical Science and Engineering, KAIST*
- 11/2022 Special Seminar – *Yonsei University College of Medicine*
- 08/2022 Young Global Leader Symposium – *Dept. of Chemical and Biomolecular Engineering, KAIST*
- 07/2022 Department Seminar – *Department of Pharmacology, SNU College of Medicine*
- 05/2022 Young Investigator Seminar Series – *Department of Biomedical Engineering, UNIST*

TEACHING EXPERIENCE

- 2022 Kaufman Teaching Certificate Program, *Massachusetts Institute of Technology*
- 2014 Teaching Assistant, Advanced RNA Biology Lab, *School of Biological Sciences, SNU*
- 2013 Teaching Assistant, Biology Lab 2, *School of Biological Sciences, SNU*
- 2012 Teaching Assistant, Biology Lab 1, *School of Biological Sciences, SNU*
- 2011 Tutor, Basic Course for Biology, *School of Biological Sciences, SNU*

TECHNICAL EXPERTISE

Computational biology

- General skills
 - Unix/Linux, AWS cloud computing environment
 - Programming languages: Python, R, Rust
 - Python scientific computing libraries: NumPy, SciPy, Pandas, etc.
 - Python machine learning frameworks: Scikit-learn, PyTorch, Tensorflow
 - Building a reproducible analysis pipeline: Nextflow, Snakemake, Docker, Singularity
 - Efficient data visualization: Matplotlib, Seaborn, Plotly

- Next-generation sequencing (NGS)
 - (sc)RNA-seq, ATAC-seq, Amp-seq, CLIP-seq, TAIL-seq, etc.
 - Statistical analysis for genome-wide, transcriptomic studies
 - Tools: Samtools, bcftools, GATK, Scanpy, Seurat, STAR, Bedtools, etc.
- Image processing
 - Concurrent programming for large-scale images
 - 2D/3D rigid and non-rigid biomedical image registration
 - Convolutional Neural Networks
 - Tools: PyTorch, Zarr, Dask, Scikit-image, OpenCV, ImageJ/Fiji, Imaris, etc.
- AI/ML
 - Biological language models: variant effect prediction, protein/RNA sequence embeddings
 - Model training and fine-tuning: PyTorch, Scikit-learn
 - Binding affinity prediction, genotoxicity risk scoring
 - Personal knowledge-base tooling: LLM-based paper ingestion and cross-corpus literature synthesis

Wet experiments

- Routine experiments
 - Molecular cloning: Gibson, Golden Gate assembly
 - Mammalian, bacterial cell culture
 - DNA and siRNA transfection
 - Quantitative RT-PCR
 - Immunoprecipitation
 - Western blotting
- RNA biology
 - Custom sequencing library preparation
 - Custom high-resolution poly(A)-tail length assay
 - Tet-On inducible reporter cell line generation
 - RNA co-immunoprecipitation
 - Lentiviral transduction
 - *In vitro* transcription
 - Northern blotting
 - Luciferase reporter assay
- Tissue clearing and expansion
 - SHIELD clearing
 - Expansion microscopy: eMAP